



Beginner's Guide

Step-by-Step Automation Guide

FileFlow Studio — Motor Visual de Automatización DAG en .NET 9



Beginner's Guide: FileFlow Studio

A Step-by-Step, Jargon-Free Guide to Organizing, Renaming, and Automating Your Files



1. What is FileFlow Studio? (In Plain English)

Imagine you have thousands of photos, songs, invoices, or movies scattered across your computer—disorganized, with messy names, or taking up way too much storage space.

Doing all of this by hand (copying, pasting, renaming one by one, unzipping...) would take days of boring work.

FileFlow Studio is like a **smart factory conveyor belt**:

1. On one end, you place a folder full of files.
2. In the middle, you connect "workstations" (called **Boxes** or **Nodes**) that perform tasks: renaming, sorting by date, compressing, converting videos, etc.
3. Your files travel through connections ("wires") from one box to another.
4. At the end, everything is clean, organized, and in its exact place in just a few seconds.

[!TIP]

You do not need to know any programming or advanced computer skills. Everything is done with your mouse: drag, drop, and connect wires.

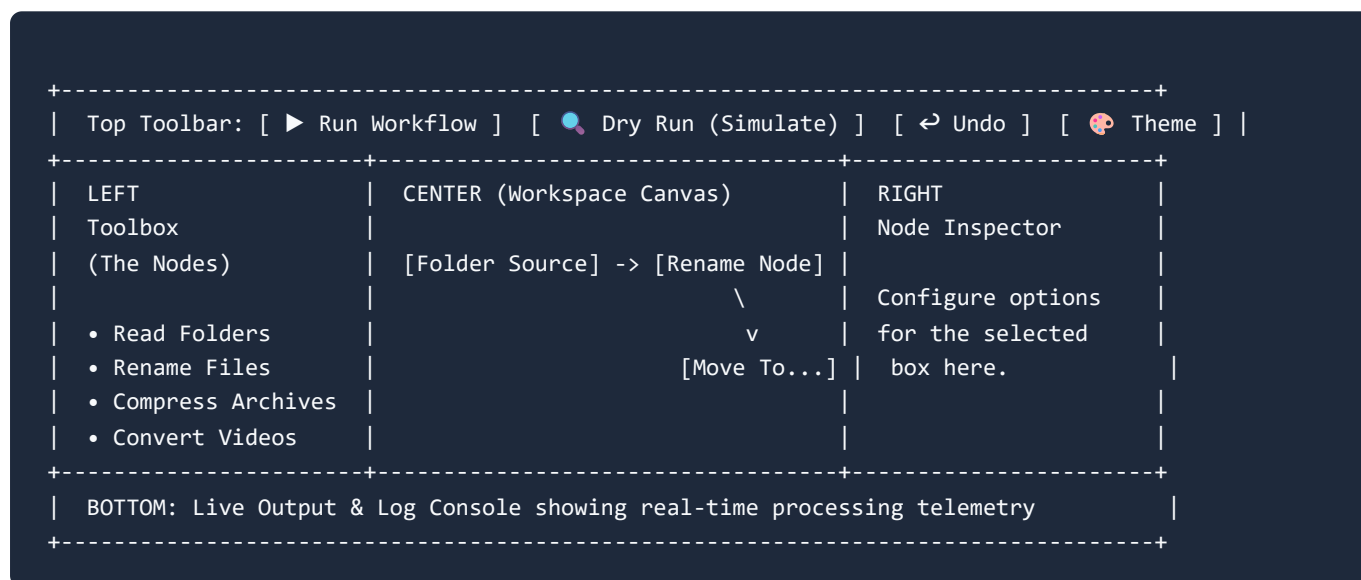
2. Peace of Mind First: The Maximum Safety Principle

Many people are afraid of using automation software in case it deletes or messes up their important photos or documents. With **FileFlow Studio**, you are 100% protected:

1. **Your original files are never altered by default:** All transformations create neat copies in new folders.
2. **The Magic "Virtual Simulation" Button (Dry Run):** You can test any workflow without moving a single real file. The app shows you an exact simulation of what it would do before you press the real button.
3. **The "Undo" Button (Ctrl+Z / Rollback):** If you run a flow and change your mind, click the **Undo** button and all files immediately return to their original names and locations.
4. **Windows Recycle Bin:** If you ever decide to delete files on purpose, the program never permanently destroys them; it sends them to the Windows Recycle Bin so you can recover them at any time.

3. Explore the Screen (The 4 Key Areas)

When you open FileFlow Studio, the screen is organized into 4 intuitive sections:



1. **Top Toolbar:** Large buttons to **Run**, **Simulate (Dry Run)**, **Pause**, **Undo**, **Save**, and toggle **Languages (English/Spanish)** or **Themes**.
2. **Left Panel (Toolbox):** Your piece catalog. Search for what you want to do (e.g., "Photos", "Rename", "Unpack") and drag it to the center.
3. **Center (Canvas / Workspace):** Where you drop boxes and connect them with wires.
4. **Right Panel (Node Inspector):** When you click any box on the canvas, its options appear here in clear language (e.g., "Where do you want to save the photos?" or "What format do you prefer?").
5. **Bottom Panel (Console):** Displays live, millisecond-by-millisecond progress for every processed file.

4. How to Connect Boxes (Editor Basics)

- **Add a box:** Click an item in the left list, hold down your mouse button, and drag it onto the center canvas.
 - **Move a box:** Click and drag the box header.
 - **Connect two boxes:** Click the circle on the right side of a box (**Output**), drag the wire, and drop it onto the left circle (**Input**) of the next box.
 - **Delete a wire or box:** Click it and press the `Delete` key on your keyboard.
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5. Four Practical Step-by-Step Recipes

Recipe 1: Rename all photos with their capture date & camera model

The Problem: Your phone or camera names photos `IMG_0042.JPG`, `DSC_9843.JPG` and you cannot tell when they were taken.

The Solution in 3 easy steps:

1. Drag the **"Folder Source"** box (`FolderSourceNode`) onto the canvas.
 - In the right panel, click **Browse...** and select your photos folder.
2. Drag the **"Advanced Renamer"** box (`AdvancedRenamerNode`).
 - Connect the output of the Folder Source to the input of the Renamer box.
 - Click the Renamer box and click the " **Method Pipeline...**" button.
 - In the Presets dropdown, choose: " **Digital Photography (EXIF Date + Camera + Counter)**".
 - You will see an instant live preview table below showing old names vs. new names!
 - Click **Save & Apply**.
3. Click the top green button "**▶ Run Workflow**".

Done! All your photos are now named `20260815_SonyA7_001.jpg`, neatly sorted by year, month, and day.

Recipe 2: Organize the Downloads Folder (Sort Videos, Photos, and Documents)

The Problem: You have 500 mixed files cluttering your "Downloads" folder.

The Solution:

1. Place a **"Folder Source"** box pointing to your `Downloads` directory.
2. Add the **"Switch / Extension Filter"** box (`SwitchCaseNode`):
 - Set up categories: `.mp4`, `.mkv`, `.avi` for Videos, `.jpg`, `.png`, `.webp` for Photos, and `.pdf`, `.docx`, `.xlsx` for Documents.

3. Connect each output port to a **"Relocate / Copy File"** box (`FileRelocatorNode`):

- Video Port $\rightarrow$ Destination: `D:\My Videos\`
- Photo Port $\rightarrow$ Destination: `D:\My Photos\`
- Document Port $\rightarrow$ Destination: `D:\My Documents\`

4. Click " **Dry Run (Simulate)**" to verify the plan in the console without touching real files.

5. Click " **Run Workflow**". In 2 seconds your Downloads folder is completely organized.

Recipe 3: Batch Extract 20 ZIP or RAR Archives at Once

The Problem: You downloaded multiple compressed archives (some with passwords) and opening them one by one is tedious.

The Solution:

1. Place a **"Folder Source"** box selecting the folder where your ZIP files are located.
 2. Add the **"Smart Archive Unpacker"** box (`SmartUnpackNode`).
 - Connect both boxes.
 - If archives have passwords, click **"Manage Passwords..."** and enter candidate keys (one per line). FileFlow Studio will automatically test each password until it finds the matching one!
 - Choose the destination folder where files should be extracted.
 3. Click " **Run Workflow**".
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Recipe 4: Compress Large Videos for WhatsApp or Mobile Sharing

The Problem: You have 1 GB camera videos and want them reduced to 50 MB for easy messaging or saving onto a small USB drive.

The Solution:

1. Place a **"Folder Source"** box pointing to your video files.
 2. Add the **"Multimedia Transcoder"** box (`MediaTranscoderNode`).
 - Connect them.
 - In the right panel, under Quick Presets, pick: **"Mobile Ultra-Compressed H.264"** or **"Fast 720p H.264 (MP4)"**.
 - Choose where to store the converted videos.
 3. Click " **Run Workflow**".
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Recipe 5: Merge Multiple Invoices or Notes into a Single PDF

The Problem: You have 10 loose PDF pages and need to send them as a single combined PDF document.

The Solution:

1. Place a **"Folder Source"** box pointing to your loose PDF files.
 2. Add the **"Merge PDFs"** box (PdfMergeNode).
 - Connect them.
 - In the right panel, enter your desired output filename (e.g., Combined_Invoices.pdf) and the destination folder.
 3. Click **"▶ Run Workflow"**.
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Recipe 6: Extract Text from Document Photos with Local AI (OCR)

The Problem: You received scanned receipts or photos of physical documents and want to extract their text automatically without typing it by hand.

The Solution:

1. Place a **"Folder Source"** box pointing to your image files.
 2. Add the **"Local OCR"** box (LocalOcrNode). It runs 100% locally on your computer with zero cloud uploads or subscription fees!
 3. Add the **"Operation Report"** box (OperationReportNode) configured for HTML or Markdown .
 4. Click **"▶ Run Workflow"** to preview all recognized text ready to copy.
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Recipe 7: Download and Upload to Remote Servers & Private Clouds (SFTP / WebDAV)

The Problem: You want to download files from an SSH server or upload backups to Nextcloud automatically.

The Solution:

1. To Download: Add the **"Network Download Hub"** box (NetworkDownloadNode), select your protocol (e.g. SFTP or WebDAV), and fill in your connection details.
 2. To Upload: Add the **"Network Upload Hub"** box (NetworkUploadNode) at the end of any pipeline to push results straight to your server.
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Recipe 8: Automatic Watchdog Mode (Process Files as They Land)

The Problem: You want new files added to your Downloads or Inbox folder to be sorted automatically in the background without manually launching FileFlow Studio every time.

The Solution:

1. Build your flow (e.g. FolderSource \$\\rightarrow\$ Filter \$\\rightarrow\$ MoveFile).
2. Click the **"🐶 Vigilante" (Watchdog)** button in the top bar.

3. The button illuminates in vibrant emerald green. You can now minimize the app: whenever a new file lands in the folder, it is immediately processed automatically.



Recipe 9: Testing Pipelines with "Mock Test Files" & Virtual File System (100% Risk-Free)

The Problem: You are building an advanced pipeline (camera-based photo renamer, archive unpacker, invoice sorter...) but do not want to risk real personal files or do not have sample files handy.

The Solution:

1. Add the **"Synthetic Data Generator"** box (`SyntheticDataSourceNode`) instead of a real folder scanner.
2. Select an out-of-the-box domain in the inspector: *"Movies"*, *"TV Series"*, *"Music"* (with realistic ID3 tags), *"Photos"* (with Sony, Canon, Nikon EXIF data), or *"Documents"* (invoices and balance sheets).
3. Want to design your own custom folders and test files? Click "  **Diseñar Conjuntos de Datos...**" in the right inspector:
 - Create custom directory trees with intuitive indented text (Fast Tree DSL) or a simple visual table.
 - You can even simulate compressed `.zip` archives with inner files!
4. Connect your destination nodes (Rename, Move, Save...) and click "► **Run Workflow**".
5. **The Magic Behind It:** Zero files are created on your actual hard drive! Everything is written safely into an in-memory **Virtual File System (VFS)**.
6. When the execution finishes, click the new "  **VFS**" button in the top bar:
 - Inspect the generated virtual folder hierarchy, operation status badges (*Saved*, *Moved*, *Renamed*), and rich metadata.
 - If satisfied, click *"Open in Explorer"* to export the sandbox or switch to real folder inputs with confidence.



The Spacebar Trick (QuickLook Instant Previewer)

Want to check how a photo turned out, read an extracted document, or review metadata tables without opening third-party tools?

1. Click on any file in the console or a card on the canvas.
 2. Press the **Spacebar**.
 3. A sleek preview window pops up showing images, text, and metadata tables on the fly! Press `Spacebar` or `Esc` again to close it.
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6. Canvas Productivity & Organization Tips

- **Sticky Notes:** Right-click an empty canvas area and choose **"Create Sticky Note"** to add colorful notes, guidelines, or reminders directly on the workspace.
 - **Group Frames (`Ctrl+G`):** Select multiple related boxes by dragging a selection rectangle and press `Ctrl+G`. They will be enclosed in an interactive group frame with a customizable title that moves all boxes together.
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7. Frequently Asked Questions (FAQ)

1. What happens if the power goes out while processing?

FileFlow Studio processes each file atomically and records checkpoints automatically. Nothing is corrupted. When you restart the app, it automatically resumes and skips already completed files.

2. How do I know when the flow is finished?

The top progress bar fills with green and the bottom console outputs a final summary: *"Workflow completed successfully! X files processed in Y seconds"*.

3. Can I save my workflows for daily use?

Yes! Click **"Save Workflow As..."** in the side drawer (or press `Ctrl+S`). You can save as many workflows as you want (e.g., *"CleanDownloads.flow"* or *"OrganizePhotos.flow"*). To reuse it, simply open the saved file and click Run or Watchdog.

A workflow saved with an older version of FileFlow Studio **brings itself up to date when you open and save it**: your wires stay where they were and the file ends up current without you doing anything else. And if a file was written by a newer version, the application warns you before overwriting it and offers to save it under another name, so nothing is ever lost.

4. What do the colored status circles on each box mean?

-  **Gray (Idle):** The box is waiting for incoming files.
 -  **Yellow / Blue (Working):** The box is actively processing an item right now.
 -  **Green (Completed):** Finished processing all items successfully.
 -  **Red (Faulted / Alert):** An issue occurred with a file (e.g., disk was full or a document was locked by Word). You can read the exact reason in the console below.
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8. Jargon Buster Glossary

Term	What it Means in Plain English
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Node / Box	A visual block that performs a specific task (rename, copy, filter, compress, etc.).
Workflow / Pipeline	The complete chain of connected boxes from start to finish.
Watchdog Mode (Vigilante)	Background listening mode that monitors folders and processes new files automatically.
Dry Run / Simulation	A safe test rehearsal: performs all calculations without modifying your real files.
Rollback / Undo	Rewinds the changes: restores your files back to their original names and locations.
State Checkpointing	Automatic progress persistence that resumes long-running tasks without re-doing finished work.
Metadata (EXIF / ID3 / Columns)	Hidden data inside a file (photo capture date, camera model, song artist, Excel columns).
Synthetic Data (Samples)	Harmless mock files generated by the system to test workflows without needing your real files.
VFS (Virtual File System)	A simulated RAM hard drive where test workflows write and organize files without touching your disk.
Portable Version	A standalone version of FileFlow Studio you can carry on a USB drive and run on any PC without installation.

Enjoy organizing and automating your files with **FileFlow Studio!** 🎉

Enjoy fast, safe, and effortless file automation with **FileFlow Studio!** 🎉